

We can then solve for x , using (1), which gives $2x = 1 - y$. Thus

$$2x = 1 - \frac{-5}{7} = \frac{7+5}{7} = \frac{12}{7}.$$

Hence

$$x = \frac{12}{2 \cdot 7} = \frac{12}{14}.$$

Our answer is therefore:

$$y = \frac{-5}{7} \quad \text{and} \quad x = \frac{12}{14}.$$

If we want x in lowest form, we can always write $x = \frac{6}{7}$, but $\frac{12}{14}$ is quite correct.

As a variation, we could also have eliminated y first. Thus we multiply the first equation by 2, leave the second unchanged, and add the equations. We get:

$$\begin{aligned} 4x + 2y &= 2, \\ 3x - 2y &= 4. \end{aligned}$$

Adding yields

$$4x + 3x = 6.$$

Thus $7x = 6$ and $x = \frac{6}{7}$, which is of course the same answer that we found above. We could then solve for y using the first equation, namely,

$$y = 1 - 2x,$$

so that

$$y = 1 - \frac{12}{7} = \frac{7-12}{7} = \frac{-5}{7}.$$

It may happen that a system of linear equations has no solutions. For instance the system

$$\begin{aligned} (3) \quad 2x - y &= 5, \\ 2x - y &= 7 \end{aligned}$$

obviously has no solution. The system

$$\begin{aligned} (4) \quad 2x - y &= 5, \\ 6x - 3y &= 7 \end{aligned}$$

has no solution either. Indeed, any solution of $6x - 3y = 7$ is also a solution of

$$2x - y = \frac{7}{3},$$

Divide the equation by 3), and it is again obvious that no simultaneous solution exists for the system of equations (4).

We do not want to overemphasize here the theory determining precisely the cases when a solution exists and when it does not. It “usually” exists, unless one has a case essentially like the examples above. Our purposes here are mainly to put you at ease with two simple equations in two unknowns, so that you have some simple approach to them. We don’t intend to overburden you or give you any worries about them. On the other hand, you may wish to do Exercises 9 and 10 to get the general criterion indicating when a solution exists. These exercises make precise our meaning of “usually”.

When you learn about coordinates, then you will see that the simultaneous equations we have been considering represent straight lines, and that finding their simultaneous solution gives the coordinates of the point of intersection of these lines. If you wish, you may look up coordinates right away, and the first section of Chapter 12 to see about this.

One final remark. Observe that our elimination procedure actually proves that if x, y are numbers satisfying the simultaneous equations, then they must have the value obtained by the method indicated. Conversely these values for x, y actually are solutions of the equations. This can be checked each time explicitly. To prove it in general is easy but requires setting up convenient notation and using general letters for the coefficients of the equation. See Exercise 11. Here we don’t want to get bogged down in abstraction. Our purpose in this section was simply to teach you a simple and efficient way of finding the solutions of a simple system of equations.

Simultaneous equations like the above can be used to solve problems which we gave in the context of one variable at the end of Chapter 1. We give an example of this.

Example. A person takes a trip and drives 8 hr, a distance of 400 mi. His average speed is 60 mph on the freeway, and 30 mph when he drives through a town. How long did the person drive through towns during his trip?

To solve this, let x be the length of time driven on freeways, and let y be the length of time driven through towns. Then

$$x + y = 8.$$

This gives us a first equation. Furthermore, the distance driven on freeways is equal to $60x$, and the distance driven through towns is equal to $30y$. Hence we get a second equation

$$60x + 30y = 400.$$

We can now solve our pair of equations, by multiplying the first by 60 and subtracting the second. We get

$$60y - 30y = 480 - 400,$$

or more simply,

$$30y = 80.$$

Therefore

$$y = \frac{80}{30} = \frac{8}{3}$$

is our numerical answer, and the person drove $\frac{8}{3}$ hr through towns. This is of course the same answer that we found when working with only one variable.

You may now wish to work out the exercises at the end of Chapter 1, §6 by means of two unknowns, which may be easier to handle the problems.

EXERCISES

Solve the following systems of equations for x and y .

1. $2x - y = 3$

$$x + y = 2$$

2. $-4x + 7y = -1$

$$x - 2y = -4$$

3. $3x + 4y = -2$

$$-2x - 3y = 1$$

4. $-3x + 2y = -1$

$$x - y = 2$$

5. $-3x + y = 0$

$$x - y = 1$$

6. $3x + 7y = 0$

$$x - y = 0$$

7. $7x - y = 2$

$$2x + 2y = 4$$

8. $-4x - 7y = 5$

$$2x + y = 6$$